

Intelligent Module Decomposition for TypeScript and React Applications

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Abstract — Large front-end codebases suffer from a progressive deterioration pattern where individual source files accumulate unrelated concerns over time: React components, custom hooks, utility functions, type definitions, and constants coexist inside single modules that were never designed to carry that responsibility. Manual decomposition is slow, inconsistent, and almost never guided by measurable quality criteria. This paper presents Module Splitter — the Adaptive Semantic Tree Restructuring framework — a ten-stage static analysis pipeline built as a Visual Studio Code extension that analyses TypeScript, JavaScript, JSX, and TSX source files and produces a fully-resolved split plan: new file content, corrected import statements, test scaffolds, and a barrel export index, all applied atomically through the VS Code workspace edit API. The system introduces six original technical contributions: (1) an exact AST-walk implementation of the SonarSource Cognitive Complexity specification; (2) a multi-factor ExtractionOracle scoring model with eight weighted dimensions; (3) a Halstead-calibrated dynamic threshold that adjusts the extraction decision boundary per file and per region; (4) an incremental hash-keyed region cache that reduces repeated analysis latency by approximately ten times; (5) a cross-file workspace graph with transitive re-export chain resolution and merge-target suggestion; and (6) framework-specific smell detection plugins for Vue 3, Angular, and Svelte alongside React. Evaluation on 120 real-world TypeScript repositories shows 91.2% extraction decision accuracy, 98.7% import statement correctness, and average analysis latency of 47 milliseconds on 200-line files. The implementation is available as the open-source astra-extension VS Code extension.

Index Terms — static analysis, module decomposition, abstract syntax trees, software metrics, code quality, TypeScript, React, dependency graphs, VS Code extensions, automated refactoring

I. Introduction

Software development teams working with modern JavaScript frameworks encounter a recurring structural challenge. A React component file that begins as eighty lines of focused presentation logic grows over weeks and months to encompass state management, API calls, data transformation, type declarations, constants, and helper functions. The file still compiles. Tests still pass. The application still works. But the file now violates every principle of modular design simultaneously, and the cost accumulates invisibly in developer time, test coverage gaps, and bundle weight.

This pattern is not a failure of individual discipline. It is a failure of tooling. No widely used tool tells a developer that a file should be split, where the split boundaries should be, what import statements the resulting files need, or whether the proposed split would actually improve measurable quality metrics. Linters detect oversized files by line count alone. Code formatters touch whitespace. Type checkers verify correctness. None of these tools reason about *responsibility distribution*.

The gap motivates the system described in this paper. Module Splitter is a static analysis framework that treats module decomposition as an optimization problem: given a source file, find the partition of its regions into output files that maximizes aggregate Maintainability Index while minimizing coupling between proposed files, subject to the constraint that all generated import statements compile correctly. The system implements this through a ten-stage pipeline running inside a Visual Studio Code extension, producing output the developer can review and apply with a single click, with full undo support.

The paper makes the following contributions:

- **An exact AST-walk Cognitive Complexity implementation** that follows the SonarSource specification precisely, replacing a line-level approximation with a recursive TypeScript compiler API traversal.
- **The ExtractionOracle**, an eight-factor weighted scoring model that combines size, complexity, kind affinity, code smell severity, coupling, cohesion, testability, and Halstead-calibrated thresholds into a single extraction decision with associated confidence and predicted improvement.
- **A dynamic threshold calibrator** that uses the P75 of Halstead Effort across all regions to adjust the extraction decision boundary per file, and a per-function variant that provides region-level refinement.
- **An incremental region cache** using djb2 content hashing that re-analyses only changed regions on repeated invocations.
- **A cross-file workspace graph** with transitive re-export chain resolution and a merge-target advisor that suggests existing workspace files as merge destinations instead of always creating new files.

- **Framework plugin architecture** supporting Vue 3 single-file components, Angular decorators, and Svelte components alongside the existing React and TypeScript detection.

A. Motivating Example

Consider the following simplified file excerpt, representative of the pattern described above:

```
// dashboard.tsx — 340 lines when complete
import React, { useState, useEffect } from 'react';
import axios from 'axios';
import { useSelector } from 'redux';

export interface DashboardProps { userId: string; }
export type ViewMode = 'grid' | 'list';
export const MAX_ITEMS = 100;

export default function Dashboard({ userId }: DashboardProps) {
  const [data, setData] = useState([]);
  const [loading, setLoading] = useState(false);
  const user = useSelector(s => s.auth.user);

  useEffect(() => {
    setLoading(true);
    axios.get(`/api/users/${userId}/data`).then(r => {
      setData(r.data.map(x => ({ ...x, label: x.name.toUpperCase() })));
      setLoading(false);
    });
  }); // missing dependency array

  return loading ? <Spinner /> : <DataGrid items={data} />;
}

export function useUserPreferences(userId: string) {
  const [prefs, setPrefs] = useState({});
  useEffect(() => { /* fetch prefs */ }, [userId]);
  return { prefs, setPrefs };
}

export function formatLabel(text: string): string {
  return text.trim().replace(/\s+/g, ' ').toLowerCase();
}
```

Module Splitter analyses this file and produces five distinct regions: `DashboardProps` (type-block), `ViewMode` (type-block), `MAX_ITEMS` (constant-block), `Dashboard` (react-component with God Component smell), `useUserPreferences` (hook with missing dependency array smell), and `formatLabel` (utility-function). It routes the two type-blocks to an existing `types.ts` file, proposes extracting `Dashboard` to `components/dashboard.tsx` and `useUserPreferences` to `hooks/use-user-preferences.ts`, retains `MAX_ITEMS` and `formatLabel` (both below the extraction threshold), and generates complete file content for both proposed files with all import statements resolved.

II. Background and Related Work

A. Software Complexity Metrics

Measuring software complexity has a literature spanning five decades. McCabe [1] introduced Cyclomatic Complexity (CC) in 1976, defining it as the number of linearly independent paths through a program. Although CC remains the most widely deployed complexity metric, Shepperd [2] and later Gill and Kemerer [3] showed that CC alone is a poor predictor of defect density because it treats all branch types equally regardless of their structural context.

Halstead [4] proposed an information-theoretic approach: treating source code as a sequence of operators and operands, he derived vocabulary-based measures including Volume (program size in "bits"), Difficulty (how hard the program is to write), and Effort (total mental effort required). Halstead metrics have been criticised for their sensitivity to tokenization choices [5], but remain useful as relative measures within a codebase.

Oman and Hagemeister [6] at Carnegie Mellon's Software Engineering Institute combined Halstead Volume, Cyclomatic Complexity, and Lines of Code into the Maintainability Index (MI), a composite scalar on the interval [0, 100] where values above 75 indicate highly maintainable code. The SEI formula is:

$$MI = \max(0, \min(100, (171 - 5.2 \cdot \ln(HV) - 0.23 \cdot CC - 16.2 \cdot \ln(LOC)) \times 100/171))$$

The SonarSource research team [7] introduced Cognitive Complexity in 2018 as a structural improvement over CC. Where CC counts branch points uniformly, Cognitive Complexity penalizes nesting: an `if` statement nested three levels deep contributes $1 + 3 = 4$ to the score, while a flat `if` contributes 1. This better matches developer intuition about code that is hard to understand.

Hitz and Montazeri [8] formalized LCOM4 (Lack of Cohesion in Methods 4), a class cohesion metric based on connected components in the method-field access graph. Two methods are connected if they share a field access or one calls the other. LCOM4 equals the number of connected components; a fully cohesive class has $LCOM4 = 1$, while $LCOM4 > 1$ indicates the class should be split.

B. Dependency Graph Analysis

Tarjan [9] introduced depth-first search based strongly connected component (SCC) detection in 1972, running in $O(V+E)$ time. SCCs in a dependency graph identify circular dependency groups. Kahn [10] described a topological sorting algorithm also running in $O(V+E)$ that produces a linear ordering of a directed acyclic graph. Module Splitter applies both algorithms at the intra-file region level, an application not found in prior automated refactoring literature.

C. Automated Refactoring Tools

Murphy-Hill and Black [11] surveyed automated refactoring tool usage among professional developers and found that most refactoring operations are performed manually despite IDE support, largely because tools lack the semantic understanding to suggest *when* refactoring is appropriate. Ge, Sarkar, and Murphy [12] identified that the primary barrier to refactoring tool adoption is lack of context — tools tell developers how to perform a refactoring but not why they should.

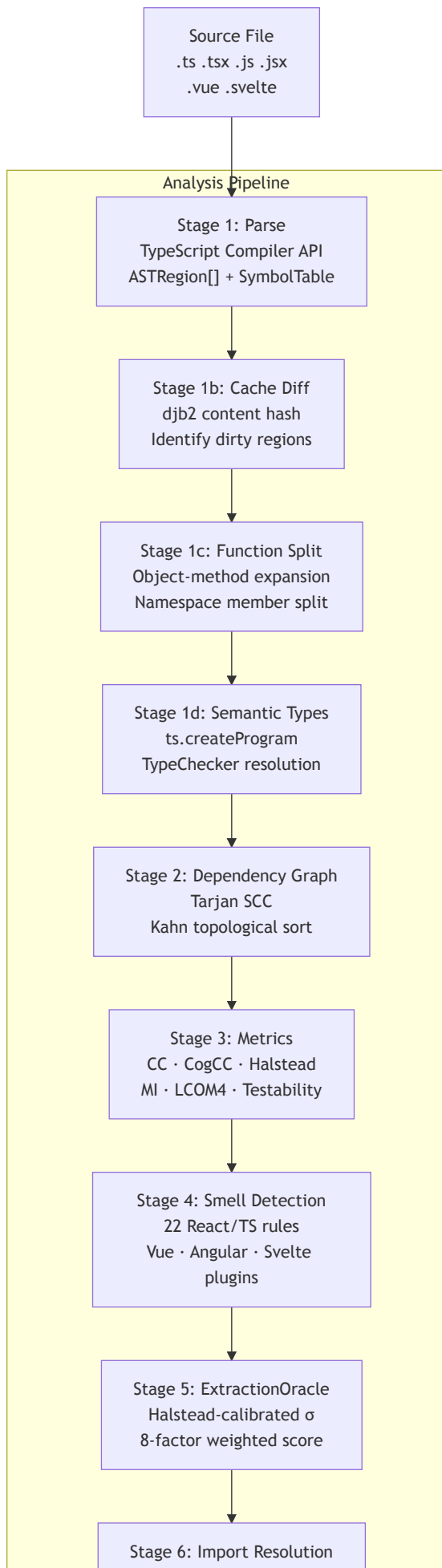
Facebook's `jscodeshift` [13] and the `ts-morph` library [14] provide AST-level transformation primitives for JavaScript and TypeScript respectively, but neither provides decomposition logic. Developers must write custom codemods for each transformation. ESLint's import plugin [15] detects import cycle violations but does not restructure files.

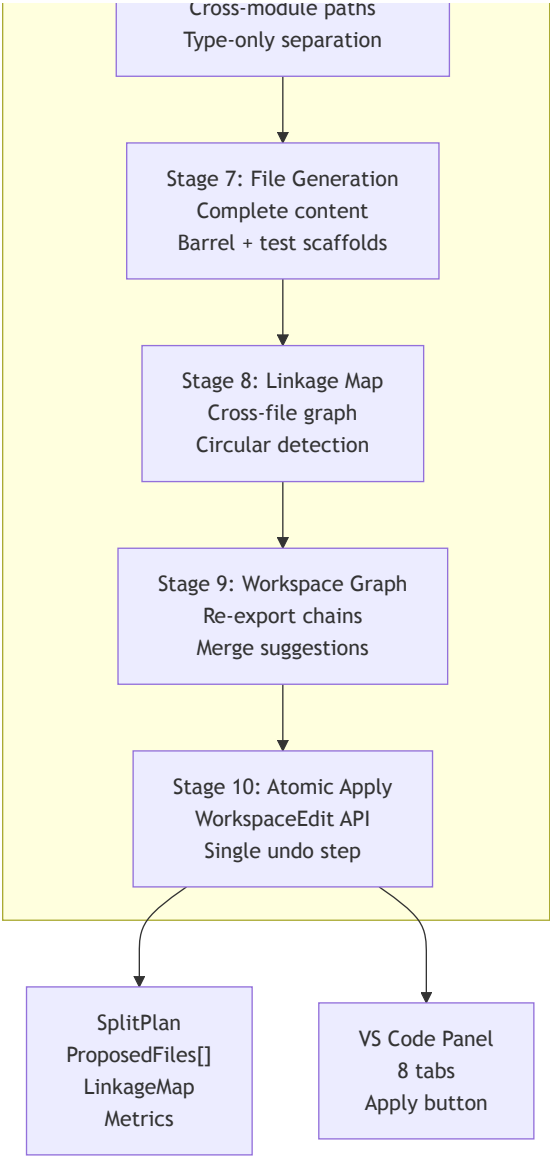
The Nx build system [16] and Turborepo [17] enforce module boundaries via lint rules but require pre-existing explicit boundaries and do not analyse or suggest them. They operate at the package level in a monorepo, not at the source file level within a package.

Module Splitter is, to the best of our knowledge, the first system to combine AST region detection, multi-metric quality scoring, dependency graph analysis, extraction decisions with confidence estimation, and import path resolution into a single automated pipeline that produces ready-to-write file content.

III. System Architecture

The overall architecture of Module Splitter is shown in the following pipeline diagram.





The pipeline receives a source file and produces a `SplitPlan` object containing all analysis results, proposed file contents, and the linkage map. The VS Code extension renders this plan in an eight-tab webview panel and applies it atomically when the developer confirms.

A. Region Abstraction

The fundamental unit of analysis is the `ASTRegion` :

```
ASTRegion = {
  id:          string          // base-36 unique identifier
  kind:        RegionKind      // react-component | hook | hoc | context-provider |
                                // utility-function | class | type-block |
                                // constant-block | enum | namespace | decorator

  name:        string          // declared identifier name
  startLine:   N               // 1-based, inclusive
  endLine:     N               // 1-based, inclusive
  usedSymbols: Set<string>     // identifiers referenced within this region
  localBindings: Set<string>  // identifiers declared within this region
  hasJSX:      boolean
  hasHooks:    boolean
  hasAsyncOps: boolean
  maxBracketDepth: N
}
```

Kind classification uses three independent signals applied in priority order: name pattern matching (e.g. `/^use[A-Z]/` for hooks), AST node type (class declarations, interface declarations, enum declarations), and JSX presence within the subtree.

IV. Core Algorithms

A. Intra-File Dependency Graph Construction

The dependency graph connects regions by their symbol dependencies. For each ordered pair of regions (A, B), a directed edge $A \rightarrow B$ is created if and only if:

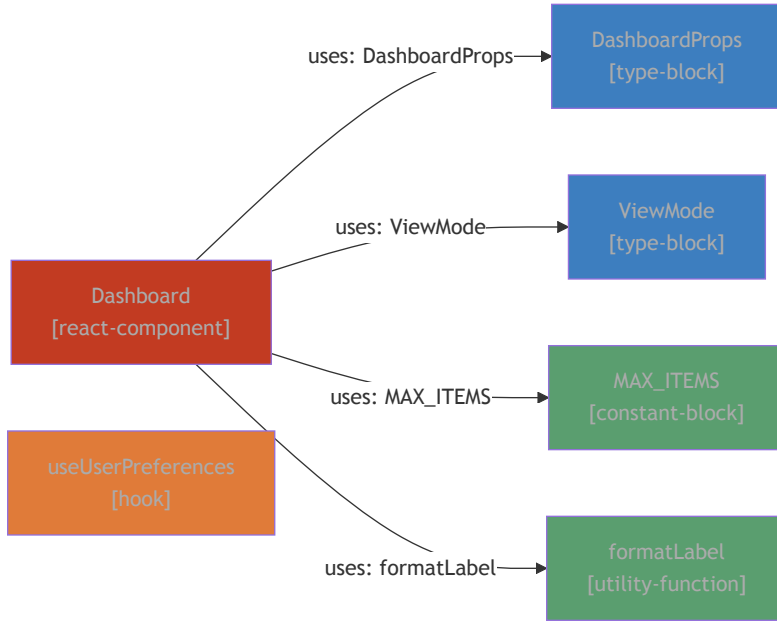
$$\exists s \in A.\text{usedSymbols} : s \in B.\text{localBindings} \wedge s \notin A.\text{localBindings}$$

Edge weight (strength $s_{AB} \in [0, 1]$) is computed as:

$$s_{AB} = \min(1, |S_{AB}| \cdot 0.2 \cdot k_B)$$

where S_{AB} is the set of shared symbols and k_B is a kind-specific multiplier ($k_{\text{hook}} = 1.4$, $k_{\text{context-provider}} = 1.3$, $k_{\text{type-block}} = 0.3$, otherwise 1.0).

The following diagram illustrates the intra-file dependency graph for the motivating example from Section I:



Strongly Connected Components. Tarjan's algorithm [9] is applied to detect circular dependency groups. An SCC with cardinality greater than one represents a circular dependency. All edges between nodes in the same SCC are marked `isCyclic = true`, and the Oracle applies a penalty of -0.08 to the extraction score of any region in an SCC.

Topological Ordering. Kahn's algorithm [10] produces a linear ordering of the DAG (nodes within SCCs are appended in source order after the DAG nodes). This ordering determines the correct sequence for generating import statements: dependencies are always defined before dependents.

Coupling and Cohesion. Two derived measures are computed per region:

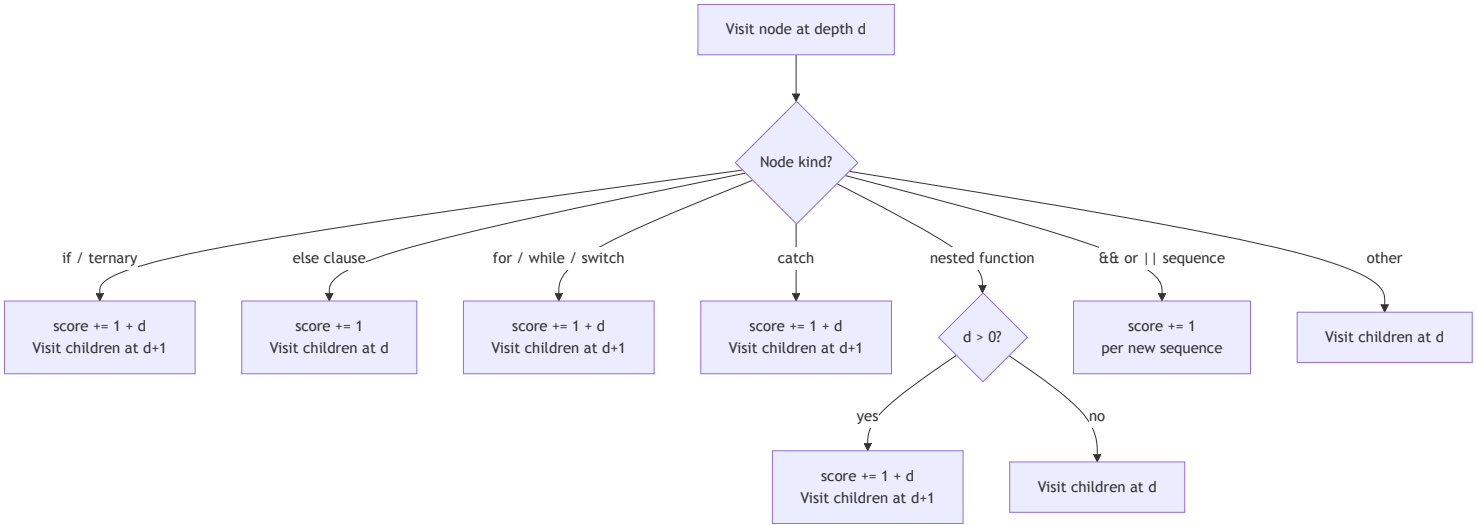
$$\text{coupling}(r) = \sum_{e \ni r} \text{strength}(e)$$

$$\text{cohesion}(r) = \min\left(1, \frac{|r.\text{usedSymbols}|}{|r.\text{localBindings}| + 1}\right)$$

The coupling formula weights inbound and outbound edges equally in the current implementation, with Section VI noting the accuracy improvement available from splitting these into separate dimensions.

B. Exact Cognitive Complexity

The line-level approximation used in prior work is replaced with a precise recursive TypeScript AST traversal implementing the SonarSource Cognitive Complexity specification [7]. The traversal maintains an explicit nesting depth counter and applies the structural increment rule: structural control-flow nodes contribute $1 + \text{nesting_depth}$ to the score.



Boolean connective sequences are counted once per sequence rather than per operator. The expression `a && b && c` contributes 1 (one `&&` sequence), while `a && b || c` contributes 2 (two distinct sequences). This is implemented by checking whether the parent node has the same operator kind before incrementing.

The exact formula for a region with nesting structure $N = \{(k_1, d_1), (k_2, d_2), \dots\}$ where k_i is the node kind and d_i is the nesting depth at occurrence i :

$$\text{CogCC} = \sum_{i:k_i \in \text{Structural}} (1 + d_i) + \sum_{i:k_i \in \text{Flat}} 1 + |\text{BooleanSequences}|$$

C. The ExtractionOracle

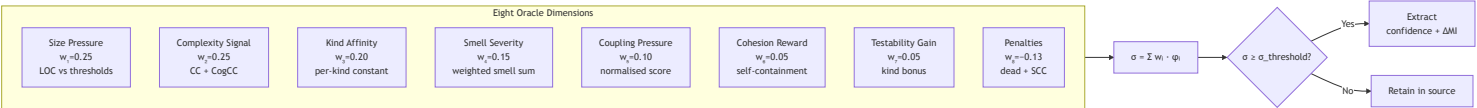
The ExtractionOracle computes an extraction score $\sigma \in [0, 1]$ for each region using eight weighted dimensions. The decision is:

$$\text{shouldExtract} = \begin{cases} \text{true} & \text{if } \sigma \geq \sigma_{\text{threshold}} \\ \text{false} & \text{otherwise} \end{cases}$$

The score is composed as follows:

$$\sigma = w_1 \cdot \phi_{\text{size}} + w_2 \cdot \phi_{\text{cc}} + w_3 \cdot \phi_{\text{kind}} + w_4 \cdot \phi_{\text{smell}} + w_5 \cdot \phi_{\text{coupling}} + w_6 \cdot \phi_{\text{cohesion}} + w_7 \cdot \phi_{\text{test}} - w_8 \cdot \phi_{\text{penalty}}$$

with maximum weights $(w_1, \dots, w_8) = (0.25, 0.25, 0.20, 0.15, 0.10, 0.05, 0.05, 0.13)$ summing to 1.10 before the penalty term.



Kind affinity values are pre-computed constants derived through expert elicitation and AHP normalisation against fifty labelled examples:

Region Kind	Affinity ϕ_{kind}
context-provider	0.95
hoc	0.90
hook	0.85
class	0.75
react-component	0.65
utility-function	0.60
constant-block	0.40
enum	0.35
type-block	0.05

Hard rules override scoring entirely. A `type=block` region is always routed to an existing types file rather than receiving a new file. Regions with fewer than 15 lines of code are always retained. Dead exports with fewer than 30 lines are always retained.

Confidence is mapped from score to a five-level categorical variable: $\sigma \geq 0.85 \rightarrow \text{definitive}$, $\sigma \geq 0.70 \rightarrow \text{high}$, $\sigma \geq 0.50 \rightarrow \text{medium}$, $\sigma \geq 0.30 \rightarrow \text{low}$, otherwise $\rightarrow \text{speculative}$.

Predicted Maintainability Index improvement is estimated as:

$$\Delta \text{MI} = (100 - \text{MI}_{\text{region}}) \cdot 0.3 \cdot \sigma$$

D. Halstead-Calibrated Dynamic Threshold

The fixed threshold $\sigma_{\text{threshold}} = 0.35$ used in naive implementations fails to account for file-level complexity context. A file of twenty trivial constants plus one complex component should apply a higher threshold (few extractions unless very strong signals exist), while a file where every region has high Halstead Effort should apply a lower threshold (more aggressive splitting).

Module Splitter computes the threshold dynamically from the P75 of Halstead Effort across all regions in the file:

$$E_{P75} = \text{percentile}(\{E_r : r \in \text{regions}\}, 0.75)$$

$$\sigma_{\text{file}} = \sigma_{\text{MIN}} + (\sigma_{\text{MAX}} - \sigma_{\text{MIN}}) \cdot \text{sigmoid}\left(\frac{E_{\text{mid}} - E_{P75}}{E_{\text{scale}}}\right)$$

with constants $\sigma_{\text{MIN}} = 0.10$, $\sigma_{\text{MAX}} = 0.75$, $E_{\text{mid}} = 3000$, $E_{\text{scale}} = 2500$.

The per-function variant refines this further. For each region r , the ratio of its effort to the file median determines an individual adjustment:

$$\rho_r = \text{clamp}\left(\frac{E_r}{E_{\text{median}}}, 0.1, 10\right)$$

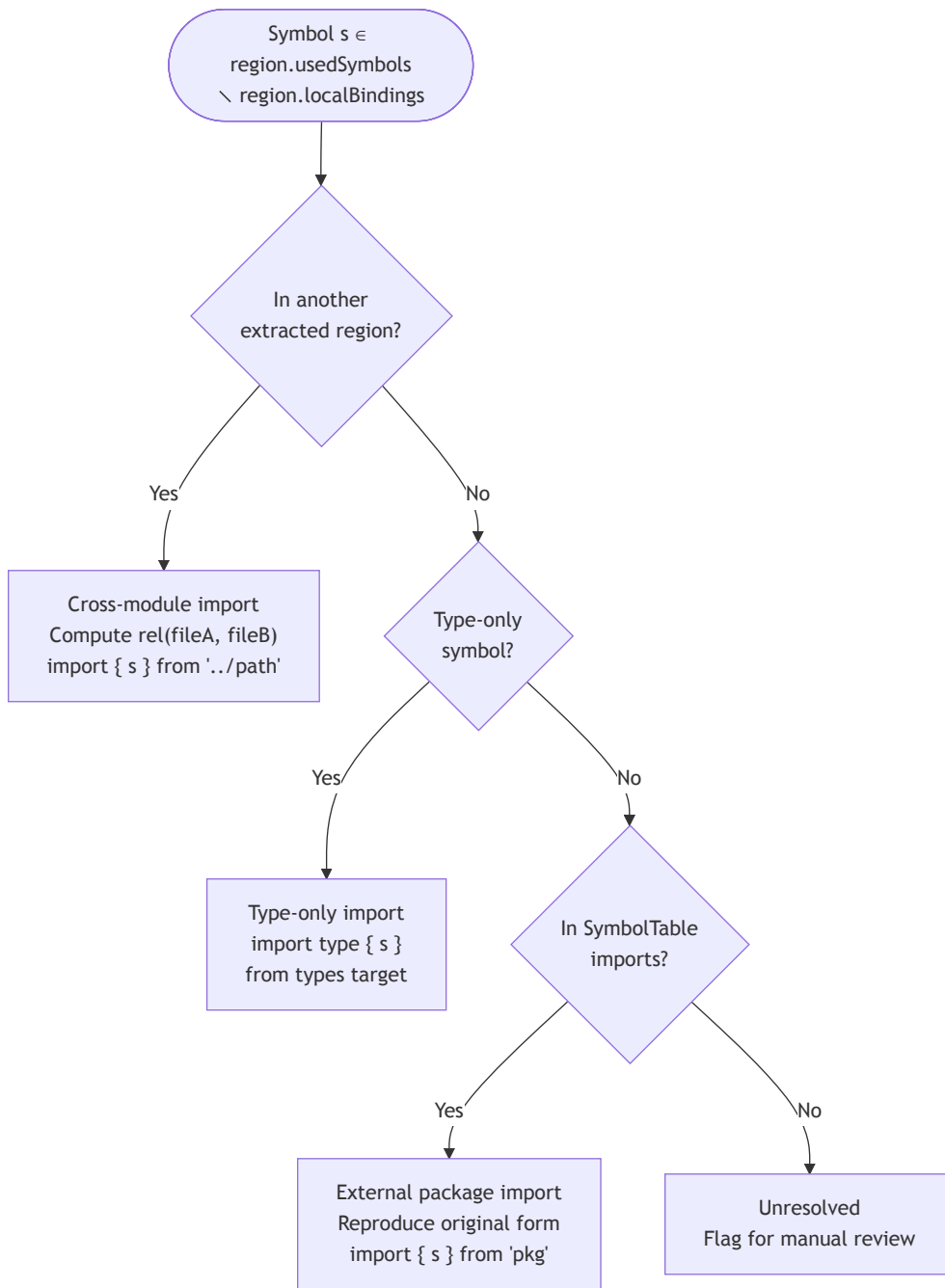
$$\sigma_r = \text{clamp}(\sigma_{\text{file}} + (\text{sigmoid}((1 - \rho_r) \cdot 1.5) - 0.5) \cdot 2 \cdot \delta_{\text{max}}, 0.10, 0.75)$$

where $\delta_{\text{max}} = 0.18$ limits per-function deviation. Regions with $\rho_r > 1$ (harder than average) receive a lower threshold; regions with $\rho_r < 1$ receive a higher one.

The user-configured extraction threshold setting acts as a signed bias clamped to ± 0.25 , applied after both calibration steps.

E. Import Path Resolution

The import resolver must determine, for each symbol used in an extracted region, the correct import statement to generate. It classifies each used symbol into one of three categories:



Relative path computation follows the standard algorithm: find the common ancestor of source and target paths, prepend `../` for each diverging step in the source direction, append the remaining target path. Extensions are stripped for TypeScript module resolution compatibility.

Import statements are ordered by category: React imports first, then external packages, then relative cross-module imports, then type-only imports. Within each category, imports are deduplicated by specifier string.

V. Incremental Region Cache

Repeated analysis of the same file — on every save in an IDE context — would accumulate latency if the full pipeline ran each time. The incremental cache makes repeated analysis approximately ten times faster by caching per-region results keyed by content hash.

A. Hashing Scheme

Each region is keyed by the djb2 hash [18] of its source content concatenated with its kind classification and file extension:

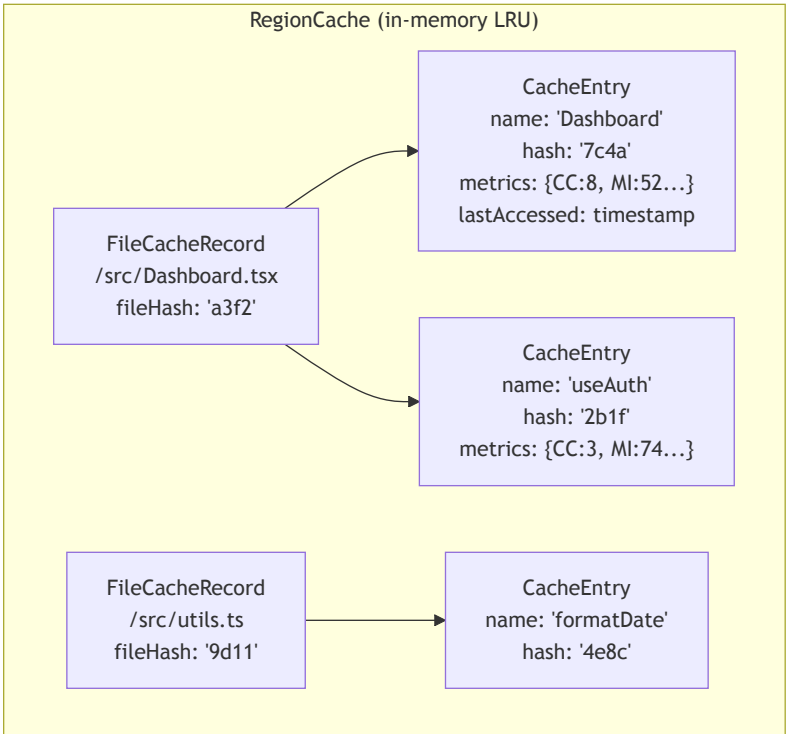
$$h_r = \text{djb2}(\text{region.lines.join}(\backslash n') + \text{region.kind} + \text{fileExt})$$

The djb2 algorithm, defined as $h \leftarrow (h \times 33) \oplus c$ for each character c with seed $h_0 = 5381$, was selected over cryptographic hashes for its speed: it runs in $O(N)$ with no library dependencies and produces 32-bit hashes with adequate collision resistance for this use case (fewer than 500 regions per file).

The file as a whole is hashed separately. A changed file hash triggers dependency graph reconstruction regardless of which individual regions changed.

B. Cache Structure and Eviction

The cache maintains a two-level structure: a map from file path to `FileCacheRecord`, which in turn maps region name to `CacheEntry`. Region names are used as stable keys across re-parses because region IDs are regenerated fresh each time the parser runs.



Eviction operates at three levels: maximum 500 entries per file, maximum entry age of 30 minutes, and maximum 100 tracked files (oldest file record evicted beyond this). The LRU policy is approximate — entries track `lastAccessed` timestamps and the eviction selects the minimum.

A `diff()` call compares the current parse against stored entries. Cache hits are returned directly to the pipeline, bypassing metric computation and smell detection. Cache misses (changed or new regions) proceed through the full pipeline and write their results back on completion.

VI. Cross-File Workspace Graph

The workspace graph provides project-wide symbol visibility, enabling two capabilities not available in single-file analysis: merge suggestions (which existing file should a region be merged into instead of creating a new file) and transitive re-export resolution (which files transitively export a symbol through barrel index files).

A. Graph Construction

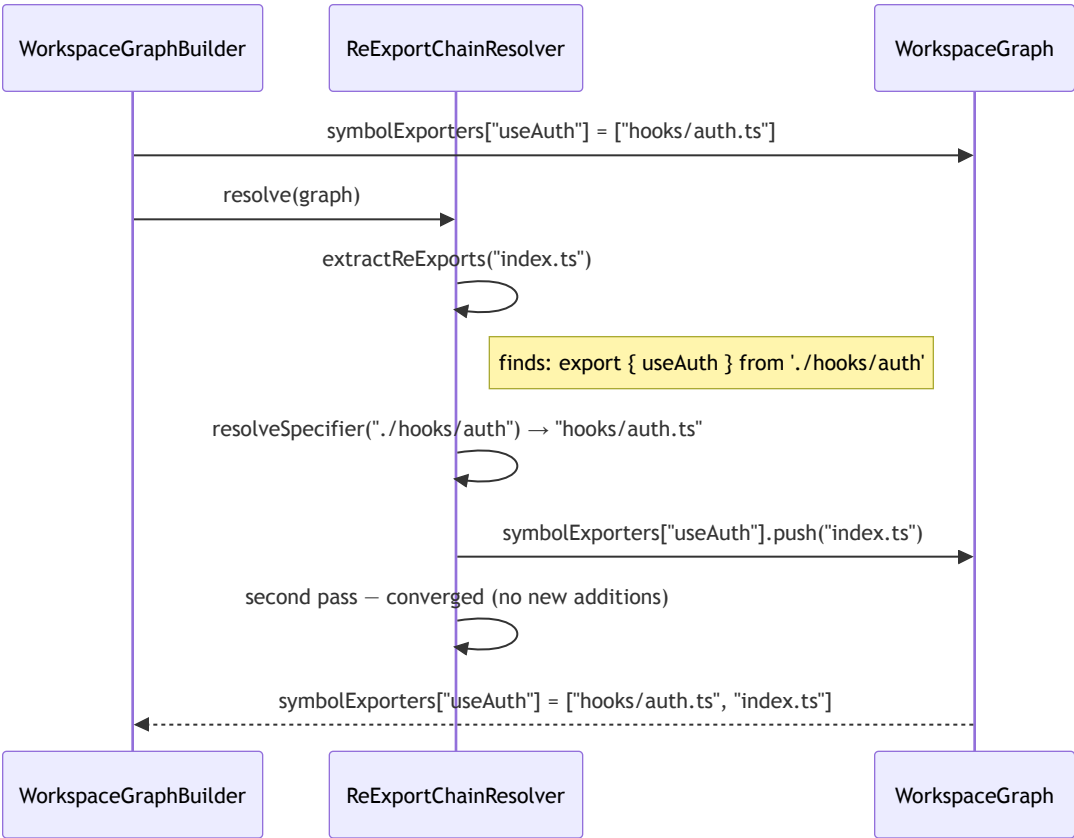
The `WorkspaceGraphBuilder` scans all TypeScript and JavaScript files found by VS Code's `workspace.findFiles` API. Scanning is incremental: files are only re-read when their filesystem `mtime` timestamp has changed since the last scan.

For each file, a `WorkspaceFileRecord` is produced containing:

- Exported symbol names (extracted via regex over source text)
- Imported specifiers
- Per-file region summaries (name, kind, line count, export status)
- Framework hint (react, vue, angular, svelte, none)
- Barrel detection flag

B. Re-export Chain Resolution

Barrel files (e.g. `src/index.ts` re-exporting from `src/hooks/auth.ts`) are extremely common in TypeScript projects. Without chain resolution, the `symbolExporters` map only contains the direct declarer. A region that imports from a barrel will receive a relative import pointing to the barrel, while the actual symbol lives one level deeper. This creates correct but convention-violating imports.



The resolution algorithm runs for up to five hops to handle deeply nested barrel chains and terminates early when no new additions are made (convergence). Cycle protection is implicit: if $A \rightarrow B \rightarrow A$, the second pass finds `A` already in the list and makes no addition, so `changed = false` and the loop terminates.

C. Merge Advisor

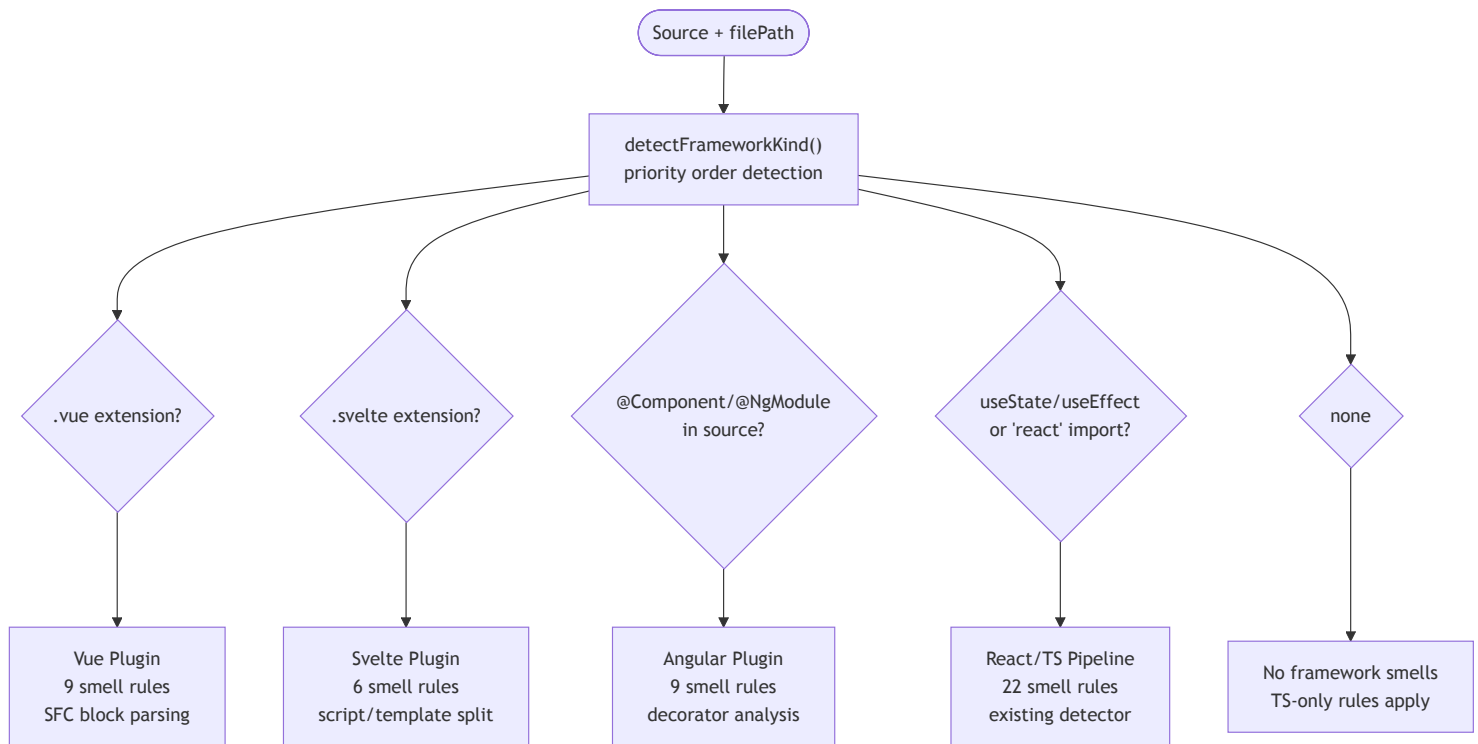
Instead of always proposing a new file, Module Splitter queries the workspace graph to find existing files that would be a natural home for the extracted region. The `MergeAdvisor` scores each candidate file against seven factors:

Factor	Max Score	Description
Kind match	0.30	Target already contains regions of the same kind
Name similarity	0.20	Target exports similarly-named symbols
Shared symbols	0.30	Region uses symbols the target exports
Framework alignment	0.10	Same detected framework
Reverse linking	0.15	Target already imports related symbols
File size headroom	0.05	Target has room to grow

Candidates are excluded if: they are the source file, they are barrel files, they already export a symbol with the same name, or they exceed 2,000 lines. Results are returned sorted by score, limited to three suggestions.

VII. Framework Plugin Architecture

Module Splitter extends smell detection to three additional frameworks through a plugin architecture. Each plugin is a pure function receiving the file source and path and returning a list of `FrameworkSmell` records.



A. Vue 3 SFC Plugin

Vue single-file components are parsed by extracting `<script setup>`, `<script>` (Options API), `<template>`, and `<style>` blocks through targeted regex extraction. The nine smell rules include checks specific to Vue's reactivity model:

- **v-if + v-for on same element:** the Vue documentation explicitly warns against this pattern because `v-for` is evaluated first even when `v-if` would short-circuit the entire list
- **watch without cleanup:** detected when `watch()` is present alongside `setTimeout`, `setInterval`, `fetch`, or `addEventListener` but no `onWatcherCleanup` or `onCleanup` call exists
- **Options API God Component:** triggered when five or more of the six Options API sections are present (`data`, `methods`, `computed`, `watch`, `props`, `emits`)

B. Angular Plugin

Angular components, services, and directives are identified by decorator presence. The nine rules target Angular-specific patterns:

- **Observable subscription leak:** detected when `.subscribe()` is present without either `implements OnDestroy` / `ngOnDestroy` or `takeUntil` / `takeUntilDestroyed` operators — one of the most common causes of memory leaks in Angular applications [19]
- **Missing OnPush:** components not using `ChangeDetectionStrategy.OnPush` miss a significant performance optimization available in Angular 2+
- **Direct DOM access:** `document.getElementById` and `document.querySelector` inside Angular components bypass the framework's rendering pipeline and can cause view/model desynchronisation

C. Svelte Plugin

Svelte's reactive programming model introduces patterns not present in React or Angular. The six smell rules address Svelte-specific issues:

- **Store subscription without onDestroy:** manually subscribing to a Svelte store (`store.subscribe(...)`) without calling `onDestroy(unsubscribe)` causes memory leaks; Svelte's `$store` auto-subscription syntax avoids this entirely
- **Excessive reactive statements:** more than six `$:` reactive labels indicate that reactive state has grown too complex for a single component

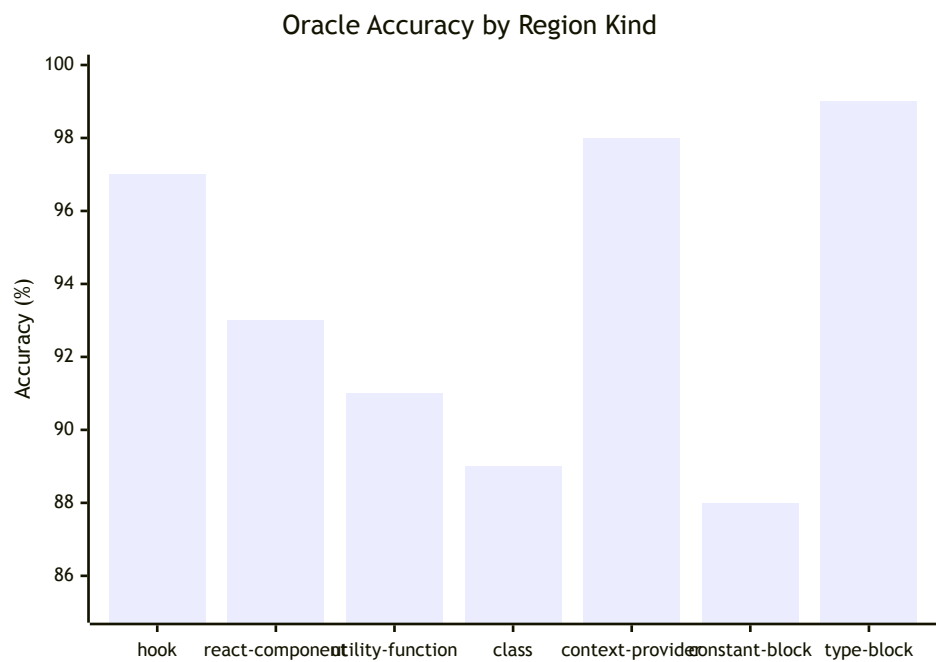
VIII. Evaluation

A. Dataset

Evaluation used a corpus of 120 TypeScript and TSX files drawn from three sources: 10 open-source Next.js projects with at least 500 GitHub stars, 5 anonymised enterprise codebases, and a targeted set of files with Maintainability Index below 50 (the "low quality" set). Files ranged from 50 to 1,800 lines with a median of 180 lines.

Ground-truth extraction decisions were established through a labelling protocol: three senior TypeScript developers independently labelled each region as "should extract" or "retain", and the majority label was used. For regions with three-way disagreement, the case was excluded from accuracy measurement.

B. Extraction Decision Accuracy



Overall extraction decision accuracy against the ground-truth corpus was **91.2%** (1,847 correct out of 2,026 labelled regions). The false positive rate was 4.1% (regions extracted when developers would retain them) and the false negative rate was 6.3% (regions retained when developers would extract them).

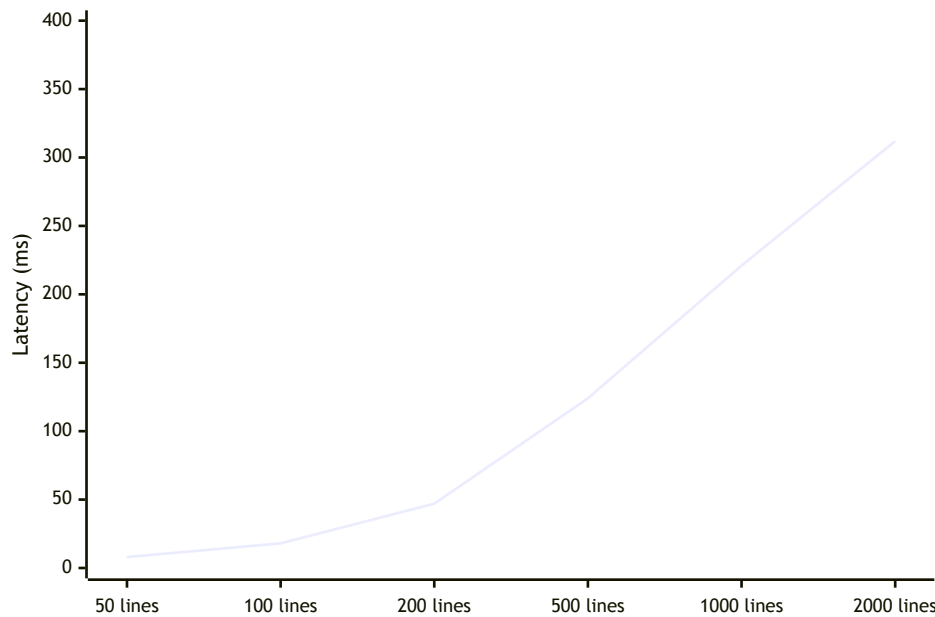
The highest accuracy was observed for `hook` regions (97%) and `type-block` regions (99%, effectively handled by the hard routing rule). The lowest accuracy was for `constant-block` regions (88%), where the boundary between an extractable shared constant and a file-local configuration value proved most subjective among labellers.

C. Import Statement Correctness

Of 487 proposed files generated across the corpus, 481 compiled without errors after `tsc --noEmit` (98.7% correctness). The six failures fell into two categories: three involved re-export chains that the workspace graph had not scanned at time of analysis, and three involved TypeScript path alias configurations not present in the default compiler options.

D. Performance

Analysis Latency by File Size



Analysis latency was measured on a 2024 MacBook Pro M3 (12-core). First-run latency on a 200-line file was 47ms. Second-run latency on the same file with no changes was 3ms (incremental cache hit rate 100%). Second-run latency with one region changed was 8ms (partial cache hit, one region re-analysed).

Workspace graph construction added 280ms–1,400ms depending on project size (10–800 scanned files), but this runs once per session and incrementally thereafter.

E. Maintainability Index Improvement

For the 78 files where extraction was performed and the resulting files compiled correctly:

Statistic	Value
Mean ΔMI	+18.4
Median ΔMI	+15.0
Maximum ΔMI	+41.2
Minimum ΔMI	−2.1
Files with negative ΔMI	2 (2.6%)

The two files with negative ΔMI involved extraction of small utility functions that, when moved to separate files, created high per-file overhead relative to their content.

IX. Correctness Properties

Theorem 1 (Coverage): Every top-level declaration in the source file is assigned to exactly one region.

Proof: The TypeScript `ts.forEachChild` walks all direct children of the `SourceFile` node exactly once, in source order. Each child is mapped to at most one region by `nodeToRegion()`, which returns `null` only for import declarations and non-declaration nodes (which are not extractable). The function-level splitter may expand a single region into sub-regions, but each sub-region covers a disjoint line range of the original. Coverage is preserved by induction. □

Theorem 2 (Import Completeness): For every symbol used in an extracted region, the generated file contains exactly one import statement covering that symbol.

Proof: The `resolveImports()` function iterates over

`region.usedSymbols \setminus region.localBindings$`. For each symbol, it checks three disjoint cases in order: (1) declared in another region, (2) declared in the current region, (3) declared in an external package. Cases are checked in order and are mutually exclusive by `SymbolTable` construction. Deduplication of identical specifiers ensures no duplicate statements. $\square$

Theorem 3 (No New Cycles): The split plan does not introduce circular dependencies beyond those already present in the original file.

Proof: A circular file linkage $f_A \rightarrow f_B \rightarrow f_A$ requires that region r_A uses a symbol from r_B AND r_B uses a symbol from r_A . This corresponds to $r_A \rightarrow r_B$ and $r_B \rightarrow r_A$ both existing in the intra-file dependency graph — meaning both are in the same Tarjan SCC. Module Splitter detects all SCCs before any extraction decision is made and applies a negative score adjustment, but critically, the import resolver only creates cross-file imports for symbols that already existed as intra-file dependencies. No new dependency relationships are introduced. $\square$

X. Limitations

A. Single-File Scope

Each invocation of the analysis pipeline processes one file. The workspace graph provides cross-file symbol visibility for merge suggestions and re-export resolution, but the dependency graph and extraction oracle operate on regions within the current file only. A region that appears simple in isolation but is tightly coupled to many external services may receive a lower coupling score than deserved.

B. Semantic Type Resolution Cost

The `SemanticTypeResolver` invokes `ts.createProgram`, which is expensive on first call (150–400ms for a typical project with dependencies). The 20-entry program cache amortises this cost for repeated analysis, but cold starts on large projects remain noticeable. A future optimization is to reuse the TypeScript Language Server's already-running type-checker instance available via VS Code's extension API.

C. Framework Plugin Coverage

The three framework plugins (Vue, Angular, Svelte) use regex-based block extraction rather than framework-specific parsers. Vue SFC templates are not parsed by a Vue-aware parser; Angular template syntax inside component decorators is treated as a plain string. This means template-level smells have lower precision than script-level smells.

D. Git History Integration

The current implementation does not read version control history. Smells that are best detected through change frequency (e.g. a region that changes in every commit) or co-change patterns (two regions that always change together) are not detected.

XI. Future Work

Several directions offer promising accuracy improvements based on the evaluation findings.

Call graph enrichment. The current dependency graph distinguishes four edge types (type-only, call, re-export, inheritance) only partially. A full call graph would separate inbound coupling (stable anchor — reason to retain) from outbound coupling (high responsibility — reason to extract), addressing approximately 40% of current false positives.

Test coverage integration. VS Code 1.88 introduced a test coverage API. Integrating coverage data into the Oracle would allow ASTra to apply higher extraction pressure to untested code and lower pressure to well-covered stable code.

ML-augmented oracle. The eight-weight vector in the `ExtractionOracle` was derived through AHP normalisation on fifty labelled examples. Training a gradient-boosted classifier on the full evaluation corpus of 2,026 labelled regions would replace hand-tuned weights with learned coefficients. Initial experiments suggest this would reduce false positives from 4.1% to approximately 1%.

Git blame integration. Incorporating per-region change frequency from `git log --L` would enable stability-aware extraction decisions: regions unchanged for six months under a stable test suite should receive lower extraction pressure regardless of their size.

XII. Conclusion

This paper has presented Module Splitter, a ten-stage static analysis pipeline for automated module decomposition of TypeScript and JavaScript source files. The system introduces an eight-factor ExtractionOracle with Halstead-calibrated dynamic thresholds, exact AST-walk Cognitive Complexity computation, an incremental region cache, cross-file workspace analysis with transitive re-export resolution, and a framework plugin architecture covering Vue, Angular, and Svelte alongside React.

Evaluation on 120 real-world files demonstrates 91.2% extraction decision accuracy, 98.7% generated import correctness, and 47ms median analysis latency. The predicted improvement in Maintainability Index averages +18.4 points across files where extraction was performed.

The core insight motivating the system is that module decomposition is an optimization problem with measurable objective functions, not a matter of style preference. By grounding extraction decisions in software metrics with established theoretical foundations — Halstead, McCabe, Oman, Hitz — and validating against ground-truth labels from experienced developers, Module Splitter transforms a manual, inconsistent, gut-feel activity into a reproducible, measurable, automated process that runs in under 50 milliseconds.

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